

Single Head Attention

We are good friend

We=[0.5,0.4,0.8,0.1]

Are=[0.2,0.6,0.32,0.7]

Good=[0.65,0.23,0.54,0.66]

Friend=[0.32,0.11,0.74,0.92]

Query:

1	1	1	1
0	0	0	0
1	1	1	1

Key:

0	0	0	0
1	1	1	1
1	1	1	1

Value:

1	1	1	1
1	1	1	1
0	0	0	0

Input Matrix:

0.5	0.4	0.8	0.1
0.2	0.6	0.32	0.7
0.65	0.23	0.54	0.66
0.32	0.11	0.74	0.92

$Q = \text{Input Matrix} * \text{Query}_{(\text{Transpose})}$

1.8	0	1.8
1.82	0	1.82
2.08	0	2.08
2.09	0	2.09

$K = \text{Input Matrix} * \text{Key}_{(\text{Transpose})}$

0	1.8	1.8
0	1.82	1.82
0	2.08	2.08
0	2.09	2.09

$V = \text{Input Matrix} * \text{Value}_{(\text{Transpose})}$

1.8	1.8	0
1.82	1.82	0
2.08	2.08	0
2.09	2.09	0

$QK = Q * K_{(\text{Transpose})}$

3.24	3.27	3.744	3.762
3.276	3.3124	3.7856	3.8038
3.744	3.7856	4.3264	4.3472
3.762	3.8038	4.3472	4.3681

S=Softmax(QK):

*softmax example for the (0,0) index: $e^{3.24}/(e^{3.24}+e^{3.27}+e^{3.744}+e^{3.762})$
=25.533/137.146=0.816

0.186	0.1918	0.308	0.313
0.185	0.192	0.308	0.314
0.1766	0.184	0.316	0.3229
0.1763	0.1838	0.3165	0.3232

Result=S*V:

Y _{we}	1.9786	1.978686	0
Y _{are}	1.97934	1.97934	0
Y _{good}	1.99951	1.999531	0
Y _{friend}	1.985664	1.985664	0

Scaled Dot Product Self Attention

We are good friend

We=[0.5,0.4,0.8,0.1]

Are=[0.2,0.6,0.32,0.7]

Good=[0.65,0.23,0.54,0.66]

Friend=[0.32,0.11,0.74,0.92]

Query:

1	1	1	1
0	0	0	0
1	1	1	1

Key:

0	0	0	0
1	1	1	1
1	1	1	1

Value:

1	1	1	1
1	1	1	1
0	0	0	0

Input Matrix:

0.5	0.4	0.8	0.1
0.2	0.6	0.32	0.7
0.65	0.23	0.54	0.66
0.32	0.11	0.74	0.92

$Q = \text{Input Matrix} * \text{Query}_{(\text{Transpose})}$

1.8	0	1.8
1.82	0	1.82
2.08	0	2.08
2.09	0	2.09

$K = \text{Input Matrix} * \text{Key}_{(\text{Transpose})}$

0	1.8	1.8
0	1.82	1.82
0	2.08	2.08
0	2.09	2.09

$V = \text{Input Matrix} * \text{Value}_{(\text{Transpose})}$

1.8	1.8	0
1.82	1.82	0
2.08	2.08	0
2.09	2.09	0

$QK = Q * K_{(\text{Transpose})}$

3.24	3.27	3.744	3.762
3.276	3.3124	3.7856	3.8038
3.744	3.7856	4.3264	4.3472
3.762	3.8038	4.3472	4.3681

Divide every index value of QK by $\sqrt{d_k}$ (where d_k Column Dimension of K which is 3 in this case).

The rest of the steps are same as that of Single Head Attention

Multihhead Attention

We are good friend

We=[0.5,0.4,0.8,0.1]

Are=[0.2,0.6,0.32,0.7]

Good=[0.65,0.23,0.54,0.66]

Friend=[0.32,0.11,0.74,0.92]

Query 1:

1	1	1	1
0	0	0	0
1	1	1	1

Key 1:

0	0	0	0
1	1	1	1
1	1	1	1

Value 1:

1	1	1	1
1	1	1	1
0	0	0	0

Query 2:

1	0	1	0
0	0	0	0
1	0	1	0

Key 2:

0	0	0	0
1	0	1	1
1	0	1	1

Value 2:

1	0	1	1
1	0	1	1
0	0	0	0

Input Matrix:

0.5	0.4	0.8	0.1
0.2	0.6	0.32	0.7
0.65	0.23	0.54	0.66
0.32	0.11	0.74	0.92

Q1=Input Matrix*Query1_(Transpose)

1.8	0	1.8
1.82	0	1.82
2.08	0	2.08
2.09	0	2.09

K1=Input Matrix*Key1_(Transpose)

0	1.8	1.8
0	1.82	1.82
0	2.08	2.08
0	2.09	2.09

V1= Input Matrix* Value1_(Transpose)

1.8	1.8	0
1.82	1.82	0
2.08	2.08	0
2.09	2.09	0

$$QK1=Q1 * K1_{(Transpose)}$$

3.24	3.27	3.744	3.762
3.276	3.3124	3.7856	3.8038
3.744	3.7856	4.3264	4.3472
3.762	3.8038	4.3472	4.3681

$$S1=Softmax(QK1)$$

*softmax example for the (0,0) index: $e^{3.24}/(e^{3.24}+e^{3.27}+e^{3.744}+e^{3.762})$
 $=25.533/137.146=0.816$

0.186	0.1918	0.308	0.313
0.185	0.192	0.308	0.314
0.1766	0.184	0.316	0.3229
0.1763	0.1838	0.3165	0.3232

$$Output1=S1*V1:$$

Y_{we}	1.9786	1.978686	0
Y_{are}	1.97934	1.97934	0
Y_{good}	1.99951	1.999531	0
Y_{friend}	1.985664	1.985664	0

$$Q2=Input\ Matrix * Query2_{(Transpose)}$$

1.3	0	1.3
0.52	0	0.52
1.19	0	1.19
1.06	0	1.06

$$K2=Input\ Matrix * Key2_{(Transpose)}$$

0	1.4	1.4
0	1.22	1.22
0	1.85	1.85
0	1.98	1.98

$$V2=Input\ Matrix * Value2_{(Transpose)}$$

1.4	1.4	0
1.22	1.22	0
1.85	1.85	0
1.98	1.98	0

$$QK2=Q2 * K2_{(Transpose)}$$

1.82	1.586	2.405	2.574
0.728	0.6344	0.962	1.0296
0.1754	0.1416	0.2997	0.3499
1.484	1.2932	1.961	2.0988

$S2 = \text{Softmax}(QK2)$:

0.170	0.134	0.3055	0.3618
0.1996	0.1817	0.2522	0.2699
0.1754	0.1416	0.2997	0.3499
0.1813	0.1498	0.2922	0.3354

$\text{Output2} = S2 * V2$:

Y_{we}	1.683	1.683	0
Y_{are}	1.502	1.502	0
Y_{good}	1.665	1.665	0
Y_{friend}	1.641	1.641	0

Concat Output1 and output2

Y_{we}	1.9786	1.978686	0	1.683	1.683	0
Y_{are}	1.97934	1.97934	0	1.502	1.502	0
Y_{good}	1.99951	1.999531	0	1.665	1.665	0
Y_{friend}	1.985664	1.985664	0	1.641	1.641	0

W_o :

1	1	1	1
0	0	0	0
1	1	1	1
0	0	0	0
1	1	1	1
0	0	0	0

$\text{Result} = \text{concat} * W_o$:

Y_{we}	3.6616	3.6616	3.6616	3.6616
Y_{are}	3.4813	3.4813	3.4813	3.4813
Y_{good}	3.66451	3.66451	3.66451	3.66451
Y_{friend}	3.6266	3.6266	3.6266	3.6266

Masked Head Attention

We are good friend

We=[0.5,0.4,0.8,0.1]

Are=[0.2,0.6,0.32,0.7]

Good=[0.65,0.23,0.54,0.66]

Friend=[0.32,0.11,0.74,0.92]

Query:

1	1	1	1
0	0	0	0
1	1	1	1

Key:

0	0	0	0
1	1	1	1
1	1	1	1

Value:

1	1	1	1
1	1	1	1
0	0	0	0

Input Matrix:

0.5	0.4	0.8	0.1
0.2	0.6	0.32	0.7
0.65	0.23	0.54	0.66
0.32	0.11	0.74	0.92

Masked Matrix: (-inf=negative infinity)

0	-inf	-inf	-inf
0	0	-inf	-inf
0	0	0	-inf
0	0	0	0

$Q = \text{Input Matrix} * \text{Query}_{(\text{Transpose})}$

1.8	0	1.8
1.82	0	1.82
2.08	0	2.08
2.09	0	2.09

$K = \text{Input Matrix} * \text{Key}_{(\text{Transpose})}$

0	1.8	1.8
0	1.82	1.82
0	2.08	2.08
0	2.09	2.09

$V = \text{Input Matrix} * \text{Value}_{(\text{Transpose})}$

1.8	1.8	0
1.82	1.82	0
2.08	2.08	0
2.09	2.09	0

$QK = Q * K_{(\text{Transpose})}$

3.24	3.27	3.744	3.762
3.276	3.3124	3.7856	3.8038
3.744	3.7856	4.3264	4.3472
3.762	3.8038	4.3472	4.3681

+

$M = QK + \text{masked matrix}$:

3.24	-inf	-inf	-inf
3.276	3.3124	-inf	-inf
3.744	3.7856	4.3264	-inf
3.762	3.8038	4.3472	4.3681

$S = \text{Softmax}(M)$:

1	0	0	0
0.4909	0.50909	0	0
0.0116	0.0316	0.0861	0
0.1763	0.1838	0.3165	0.3232

$\text{Result} = S * V$:

Y_{we}	1.8	1.8	0
Y_{are}	1.810	1.810	0
Y_{good}	0.257	0.257	0
Y_{friend}	1.3101	1.3101	0

Cross Head Attention

We are good friend

Input:

We=[0.5,0.4,0.8,0.1]

Are=[0.2,0.6,0.32,0.7]

Good=[0.65,0.23,0.54,0.66]

Friend=[0.32,0.11,0.74,0.92]

Target:

[0.7,0.3,0.5,0.2]= ہم

[0.86,0.9,0.22,0.44]= ہیں

[0.17,0.45,0.32,0.77]= اچھے

[0.61,0.89,0.47,0.24]= دوست

Query:

1	1	1	1
0	0	0	0
1	1	1	1

Key:

0	0	0	0
1	1	1	1
1	1	1	1

Value:

1	1	1	1
1	1	1	1
0	0	0	0

Input Matrix:

0.5	0.4	0.8	0.1
0.2	0.6	0.32	0.7
0.65	0.23	0.54	0.66
0.32	0.11	0.74	0.92

Target Matrix:

0.7	0.3	0.5	0.2
0.86	0.9	0.22	0.44
0.17	0.45	0.32	0.77
0.61	0.89	0.47	0.24

$K = \text{Input Matrix} * \text{Key}_{(\text{Transpose})}$

0	1.8	1.8
0	1.82	1.82
0	2.08	2.08
0	2.09	2.09

$V = \text{Input Matrix} * \text{Value}_{(\text{Transpose})}$

1.8	1.8	0
1.82	1.82	0
2.08	2.08	0
2.09	2.09	0

$Q = \text{Target Matrix} * \text{Query}_{(\text{Transpose})}$

1.7	0	1.7
2.42	0	2.42
1.71	0	1.71
2.21	0	2.21

$QK = Q * K_{(\text{Transposed})}$

3.06	3.094	3.536	3.553
4.356	4.4044	5.0336	5.0578
3.078	3.1122	3.5568	3.5739
3.978	4.0222	4.5968	4.6189

S=Softmax(QK):

0.187	0.194	0.302	0.307
0.165	0.173	0.3255	0.3335
0.187	0.193	0.302	0.307
0.171	0.179	0.319	0.326

Result=S*V

ہم	1.9594	1.9594	0
ہیں	1.9859	1.9859	0
اچھے	1.9576	1.9576	0
دوست	1.9784	1.9784	0